Robot PU

The Playful, Programmable micro:bit Robot

robotgyms.com/pu

Robot PU with gamepad remote control

Walk. Dance. Talk. Navigate. Code.

A complete, classroom-ready robot built on the **BBC micro:bit V2**. From elementary classrooms to high-school robotics clubs, Robot PU turns abstract STEM concepts into hands-on, creative projects. Two ready-to-use software options are available: a beginner track for younger students and an advanced track with more sensors, actions, and AI-camera projects. Add the optional **Smart Hat** to unlock face tracking, robot soccer, and mapping.

What's in the Kit

- Robot PU body (pre-built & upgradable)
- 2 × micro:bit compatible boards
- Gamepad for radio remote control
- Printed manual & quick-start guide
- Access to 60+ online tutorials, games & projects

Three Coding Paths

Blocks	Drag-and-drop MakeCode blocks
JS/TS	Typed JavaScript / TypeScript
Python	Python package (NovaSeq/RobotPu)

Optional: Smart Hat for AI & Soccer

Adds AI camera, voice commands, mapping, face detection, object tracking, and robot soccer. *Sold separately.*

Move & Express

Walk, turn, step sideways, kick, jump, and rest. Dance to music. Talk and sing with a friendly robot voice. Blinking LED eyes show mood and personality.

Explore & Navigate

Roams around obstacles, solves simple mazes, follows a line or path, and can trail a leader. Great for teaching problem solving and spatial reasoning.

Sense & Balance

Uses distance and tilt sensors to stay upright, avoid bumps, and react to sounds. Self-balancing motion keeps it steady on the desk or floor.

Teamwork & AI

Robots can chat by radio, sing together, and play team games. Add the Smart Hat camera to try face recognition, object tracking, and robot soccer.

Amazon: `amazon.com/dp/B0DR8RGVXN`
Website: `robotgyms.com/pu`
YouTube: The Story of PU (`youtube.com/@TheStoryofPu-yw8tr`)
Open Source: `github.com/robotgyms/pxt-robotpu`
MakeCode: `makecode.microbit.org` — add extension `robotgyms/pxt-robotpu`

Meet PU!

Ages 8–10 Beginner • Ages 11–13 Intermediate • Ages 14–17 Advanced • Difficulty: ★ Beginner ★★ Intermediate ★★★ Advanced

Setup & Basics

- Project skeleton ★
- Hello world demo ★
- JavaScript / TypeScript quick start ★★

Build & Calibration

- Add arms ★
- Servo trim calibration ★
- Material specifications ★

Robot Hardware & Behaviors

- How Robot PU moves ★
- Robot actions ★
- Robot observation ★★
- Don't bonk ★★

Music, Speech & Expression

- Music + beat-driven behaviors ★
- Dance choreography ★
- Music library utilities ★★
- Emotion state to expression ★★
- RoboVoice melodic speech ★★
- Talk content patterns ★★
- Synchronized singing ★★

Fun Projects

- Kungfu pose sequence ★★
- Learn to skate ★★★

Communication & Control

- Remote control ★
- Gamepad patterns ★
- Peer chat over radio ★★

Programming Structure

- Event loop ★★
- Custom events & handlers ★★
- State machines ★★
- State switch (setMode) ★★
- Object-oriented design ★★

Navigation & Mapping

- Sonar operator ★★
- Maze solving ★★
- Autopilot navigation ★★
- 2D mapping concepts ★★
- Submarine sonar project ★★★
- Leader-following decision engine ★★★
- Occupancy grid ★★★
- Shared mapping ★★★
- Path planning ★★★
- Path following ★★★
- Local planner ★★★

Advanced Systems

- Robots as different roles ★★
- Sharing messages between robots ★★
- Move commands ★★
- Sense the surroundings ★★
- Health checks ★★
- Activity logging ★★★
- Teacher dashboard ★★★
- Safety stop + E-stop ★★★
- Recovering from faults ★★★
- Behavior plans ★★★
- Mission plans ★★★
- Sharing tasks ★★★
- Team formations ★★★

Math & Stability

- How math makes the robot move ★★★
- Self-correcting motion ★★★
- Smoothing sensor readings ★★★
- Reliable direction sensing ★★★
- Balance on two legs ★★★
- Balance on uneven ground ★★★

AI Camera & Robot Soccer

- Smart Hat AI camera ★★★
- Find and follow a face ★★★
- Read smart-camera messages ★★★
- Track where the robot goes ★★★
- Follow a soccer ball ★★★
- Build a local map ★★★
- Keep a 1 m social distance ★★★
